BANKING CUSTOMER
DATA ANALYSIS PROJECT

Exploratory Data Analysis & Business Intelligence Report

Dataset: Banking.csv (3,000 customers | 25 features)

Tools: Python (Pandas, Seaborn, Matplotlib) | Power BI

Deliverables: Jupyter Notebook | Power BI Dashboard

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Data Analytics Portfolio Project

1. Business Problem Statement

Banks manage thousands of diverse customers with varying financial behaviors, income levels, risk profiles, and loyalty standings. Without a structured understanding of these customer segments, it becomes difficult to make informed strategic decisions around product offerings, risk management, and customer retention.

1.1 Core Business Challenges

- Identifying high-value customers who contribute disproportionately to revenue.
- Detecting customers at risk of churn or financial distress.
- Optimizing product offerings such as loans, credit cards, and business lending.
- Tailoring fee structures and loyalty programs to retain profitable segments.
- Assessing portfolio risk across different demographics and geographies.

1.2 Research Questions

This project is structured around answering the following key business questions:

- What is the financial profile of the bank's customer base in terms of income, savings, loans, and credit usage?
- How do loyalty tier and fee structure vary across customer demographics (nationality, occupation, age)?
- What correlations exist between financial variables like income, deposits, loans, and business lending?
- Which customer segments carry the highest risk weighting, and what does that mean for the bank's risk exposure?
- How can the bank better segment customers to personalize financial products and improve loyalty outcomes?

By conducting a thorough Exploratory Data Analysis (EDA) and building an interactive Power BI dashboard, this project provides data-driven insights to help bank stakeholders make informed decisions.

2. Project Overview

Attribute	Detail
Domain	Banking / Financial Services
Project Type	Exploratory Data Analysis (EDA) + Business Intelligence Dashboard

Dataset Size	3,000 customers, 25 features
Analysis Tool	Python — Pandas, NumPy, Seaborn, Matplotlib
Dashboard Tool	Microsoft Power BI
Deliverables	Jupyter Notebook (EDA) + Power BI Dashboard (.pbix)
Project Status	Completed

3. Dataset Description

The dataset contains 3,000 banking customer records with 25 features covering demographics, financial holdings, product usage, and loyalty classification.

3.1 Feature Categories

Category	Columns
Identity	Client ID, Name
Demographics	Age, Nationality, Location ID, GenderId
Occupation & Contact	Occupation, Banking Contact, BRId, IAId
Financial Products	Amount of Credit Cards, Credit Card Balance, Bank Loans, Bank Deposits, Checking Accounts, Saving Accounts, Foreign Currency Account, Business Lending, Superannuation Savings
Classification	Fee Structure, Loyalty Classification, Risk Weighting, Properties Owned
Income	Estimated Income
Timeline	Joined Bank

3.2 Key Statistics

Metric	Value
Total Customers	3,000
Total Features	25
Age Range	17 – 85 years (Mean: ~51 years)
Estimated Income Range	~₹15,919 – ₹5,22,330 (Mean: ₹1,71,305)
Average Business Lending	~₹8,66,760
Credit Cards (1/2/3)	64% / 25.6% / 10.4%

3.3 Categorical Distributions

Variable	Distribution
Loyalty Classification	Jade (44.4%) > Silver (25.6%) > Gold (19.5%) > Platinum (10.6%)
Fee Structure	High (49.2%) > Mid (32.1%) > Low (18.7%)
Nationality	European (43.6%) > Asian (25.1%) > American (16.9%) > Australian (8.5%) > African (5.9%)
Risk Weighting	Level 2 (40.7%) is most common; Level 5 (5.3%) least common
Properties Owned	Roughly uniform across 0, 1, 2, and 3 properties

4. Exploratory Data Analysis (EDA)

The full EDA is documented in final_banking_EDA.ipynb and covers the following analytical stages:

4.1 Data Loading & Initial Inspection

- Loaded the dataset using Pandas and inspected shape: (3000, 25).
- Verified data types — a mix of object, int64, and float64 columns.
- Reviewed descriptive statistics using .info() and .describe().
- Confirmed no column had structural issues; ID columns (BRId, GenderId, IAId) were treated as categorical identifiers.

4.2 Feature Engineering

- Created a new Income Band feature by binning Estimated Income into Low (< ₹1,00,000), Medium (₹1,00,000 – ₹3,00,000), and High (> ₹3,00,000).
- This derived variable enabled richer segmentation across loyalty, occupation, and nationality dimensions.

4.3 Univariate Analysis

- Plotted count distributions for all categorical features using Seaborn countplot.
- Generated KDE-overlaid histograms for 8 numerical features: Age, Estimated Income, Superannuation Savings, Credit Card Balance, Bank Loans, Bank Deposits, Saving Accounts, and Business Lending.

4.4 Bivariate Analysis

- Used Nationality as a hue variable across all categorical features to reveal cross-demographic behavioral patterns.
- Compared financial product usage across Loyalty Classification and Fee Structure tiers.

4.5 Correlation Analysis

- Generated a correlation heatmap for 8 core financial variables using Seaborn heatmap with the crest color palette.
- Identified strong, moderate, and negligible correlations guiding product strategy recommendations.

5. Key Insights & Findings

5.1 Correlation Analysis Results

Variable Pair	r Value	Strength	Business Insight
Bank Deposits ↔ Saving Accounts	0.75	Strong	Deposit-heavy customers also maintain large savings accounts
Business Lending ↔ Bank Deposits	0.44	Moderate	Business borrowers tend to also deposit more
Business Lending ↔ Bank Loans	0.42	Moderate	Heavy product users cluster together
Credit Card Balance ↔ Bank Deposits	0.38	Moderate	Higher card users also deposit more
Estimated Income ↔ Superannuation Savings	0.37	Moderate	Higher earners save more for retirement
Estimated Income ↔ Bank Loans	0.33	Moderate	Higher income customers take more loans
Estimated Income ↔ Business Lending	0.33	Moderate	Wealthier customers use business credit more
Age ↔ All Financial Variables	~0.00	Negligible	Age is not a predictor of financial activity

Key Takeaway:

Age is not a meaningful segmentation variable for this bank's customers. Financial behavior is much better explained by income level, deposit volume, and degree of product engagement.

5.2 Loyalty & Fee Structure Observations

- Jade loyalty tier dominates the customer base at 44.4% — representing a significant upsell opportunity to Silver, Gold, or Platinum tiers.
- High fee structure customers form the largest segment at 49.2%, indicating the bank is positioned as a premium-priced institution.
- No single nationality is uniformly over-represented in premium loyalty tiers — suggesting income and product usage are stronger loyalty drivers than ethnicity.

5.3 Product Usage Patterns

- 64% of customers hold only 1 credit card — clear headroom for cross-selling additional credit products.
- Business Lending varies widely (₹0 – ₹38.26 lakh), pointing to very different customer business engagement levels.
- Foreign Currency Accounts are a relatively rare product — a niche upsell opportunity for high-income or internationally-oriented customers.

5.4 Risk Profile Insights

- Risk Level 2 is most common (40.7%), indicating a moderately low-risk portfolio overall.
- Fewer than 6% of customers sit at Risk Level 5 (highest risk), limiting the bank's exposure to high-risk defaults.
- Risk weighting can potentially be cross-referenced with income bands for smarter loan approval workflows.

6. Power BI Dashboard

The Power BI dashboard (final_banking_dashboard.pbix) delivers an interactive, visual summary of the analysis with the following components:

Dashboard Component	Description
Customer Demographics Overview	Age distribution, nationality breakdown, gender split
Financial Portfolio Summary	Income bands, deposits, loan values, and savings comparisons
Loyalty & Fee Structure Analysis	Loyalty tier distribution by nationality and occupation
Risk Profiling	Risk weighting distribution and correlation with financial holdings
Product Usage Analysis	Credit card counts, foreign currency accounts, business lending trends
Interactive Filters/Slicers	Dynamic filtering by nationality, occupation, loyalty tier, and fee structure

To view the dashboard: Open final_banking_dashboard.pbix in Microsoft Power BI Desktop (free download available at powerbi.microsoft.com).

7. Conclusions & Recommendations

7.1 Conclusions

- The bank's customer base is financially diverse but skews toward moderate-to-high income earners with consistent deposit behavior.
- Income and product engagement — not age — are the primary drivers of financial activity across the customer base.
- The strong deposit-savings correlation (0.75) suggests a loyal, savings-oriented core segment that may respond well to wealth management products.
- The majority of customers in the Jade loyalty tier represent a large addressable segment for loyalty program upgrades.
- The bank's risk portfolio is generally healthy, with most customers at Risk Level 1–2.

7.2 Strategic Recommendations

Recommendation	Rationale
Cross-sell business lending to high-deposit customers	Strong correlation (0.44) between deposits and business lending usage
Upgrade Jade-tier customers with targeted loyalty incentives	44% sit at the lowest loyalty tier — large addressable upsell potential
Offer retirement/superannuation products to high-income customers	Moderate income-superannuation correlation of 0.37
Launch credit card bundle campaigns for single-card holders	64% of customers hold only 1 credit card — clear cross-sell opportunity
Develop risk-adjusted product eligibility tiers	Align product offerings with Risk Weighting scores (1–5)
Nationality-targeted marketing campaigns	European (43.6%) and Asian (25.1%) are the two dominant segments

7.3 Future Scope

- Build predictive models for customer churn using loyalty tier transitions.
- Apply clustering (K-Means / Hierarchical) to derive data-driven customer segments.
- Develop a credit risk scoring model using Risk Weighting, Income Band, and Loan data.
- Integrate time-series analysis of the 'Joined Bank' field to study customer lifetime value trends.